

# Assignment No.3

## VPC Design and Subnetting

### AWS Cloud Networking

#### Objective:

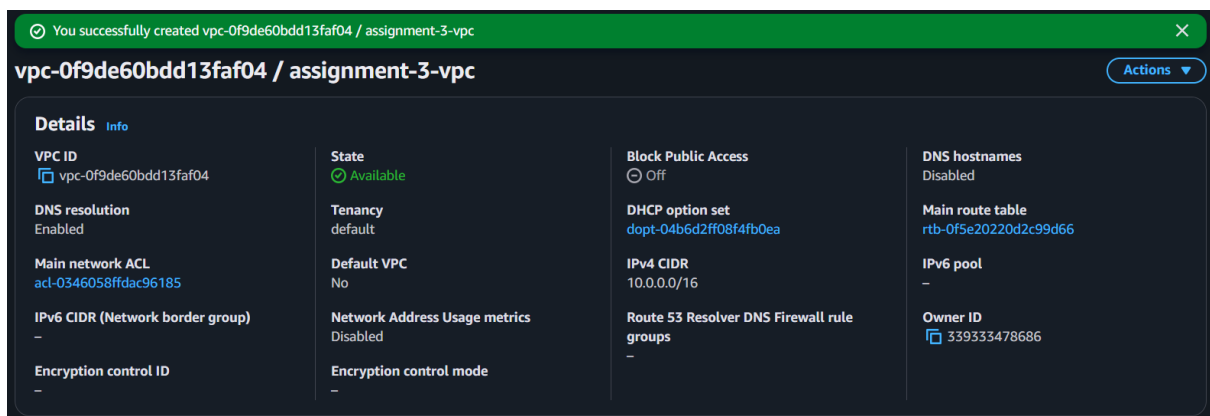
To design and configure a custom AWS VPC with public and private subnets, internet gateway, route tables, and subnet associations for secure cloud networking.

#### Services Used -

- AWS VPC
- Internet Gateway
- Route Tables
- Subnets

#### Step 1: Create Custom VPC

A custom VPC named assignment-3-vpc was created with CIDR block 10.0.0.0/16. The VPC provides an isolated virtual network environment inside AWS where cloud resources can be launched securely.



## Step 2: Create Public and Private Subnets

Four subnets were created inside the VPC across two availability zones.

### Public Subnet AZ-1:

10.0.1.0/24

The screenshot displays the AWS Management Console interface for a subnet. At the top, a green notification bar states: "You have successfully created 1 subnet: subnet-0eb710d22440747a7". Below this, the "Subnets (1/1)" section shows a table with one entry: "public-subnet-az1" with ID "subnet-0eb710d22440747a7", state "Available", VPC "vpc-0f9de60bdd13faf04", and CIDR "10.0.1.0/24". The "Details" tab is selected, showing the following information:

- Subnet ID:** subnet-0eb710d22440747a7
- Subnet ARN:** arn:aws:ec2:ap-south-1:3393334786:subnet/subnet-0eb710d22440747a7
- State:** Available
- Block Public Access:** Off
- IPv4 CIDR:** 10.0.1.0/24
- Available IPv4 addresses:** 251
- IPv6 CIDR:** -
- IPv6 CIDR association ID:** -
- Availability Zone:** ap-south-1 (ap-south-1a)
- Network border group:** ap-south-1
- VPC:** vpc-0f9de60bdd13faf04 | assignment-3-vpc
- Route table:** -
- Network ACL:** -
- Default subnet:** No
- Auto-assign public IPv4 address:** No
- Auto-assign IPv6 address:** No
- Auto-assign customer-owned IPv4 address:** No
- IPv4 CIDR reservations:** -

### Public Subnet AZ-2:

10.0.2.0/24

The screenshot displays the AWS Management Console interface for a subnet. At the top, a green notification bar states: "You have successfully created 1 subnet: subnet-0e1d1ec8939974380". Below this, the "Subnets (1/1)" section shows a table with one entry: "public-subnet-az2" with ID "subnet-0e1d1ec8939974380", state "Available", VPC "vpc-0f9de60bdd13faf04", and CIDR "10.0.2.0/24". The "Details" tab is selected, showing the following information:

- Subnet ID:** subnet-0e1d1ec8939974380
- Subnet ARN:** arn:aws:ec2:ap-south-1:3393334786:subnet/subnet-0e1d1ec8939974380
- State:** Available
- Block Public Access:** Off
- IPv4 CIDR:** 10.0.2.0/24
- Available IPv4 addresses:** 251
- IPv6 CIDR:** -
- IPv6 CIDR association ID:** -
- Availability Zone:** ap-south-1 (ap-south-1b)
- Network border group:** ap-south-1
- VPC:** vpc-0f9de60bdd13faf04 | assignment-3-vpc
- Route table:** -
- Network ACL:** -
- Default subnet:** No
- Auto-assign public IPv4 address:** No
- Auto-assign IPv6 address:** No
- Auto-assign customer-owned IPv4 address:** No
- Customer-owned IPv4 pool:** -
- Outpost ID:** -
- IPv4 CIDR reservations:** -

## Private Subnet AZ-1:

10.0.11.0/24

The screenshot displays the AWS Management Console interface for a subnet. At the top, a green banner indicates the successful creation of subnet-0ada64a7923c704de. Below this, the 'Subnets (1/1)' section shows a table with one entry: 'private-subnet-az1' with ID 'subnet-0ada64a7923c704de', state 'Available', VPC 'vpc-0f9de60bdd13faf04', and IPv4 CIDR '10.0.11.0/24'. The details panel for this subnet shows the following information:

- Subnet ID:** subnet-0ada64a7923c704de
- Subnet ARN:** arn:aws:ec2:ap-south-1:3393334786:86:subnet/subnet-0ada64a7923c704de
- State:** Available
- Block Public Access:** Off
- IPv4 CIDR:** 10.0.11.0/24
- Available IPv4 addresses:** 251
- IPv6 CIDR:** -
- Availability Zone:** ap-south-1 (ap-south-1a)
- Network border group:** ap-south-1
- VPC:** vpc-0f9de60bdd13faf04 | assignment-3-vpc
- Route table:** -
- Network ACL:** -
- Default subnet:** No
- Auto-assign public IPv4 address:** No
- Auto-assign customer-owned IPv4 address:** No
- Customer-owned IPv4 pool:** -
- Auto-assign IPv6 address:** No
- IPv4 CIDR reservations:** -
- Resource name DNS A record:** -

## Private Subnet AZ-2:

10.0.12.0/24

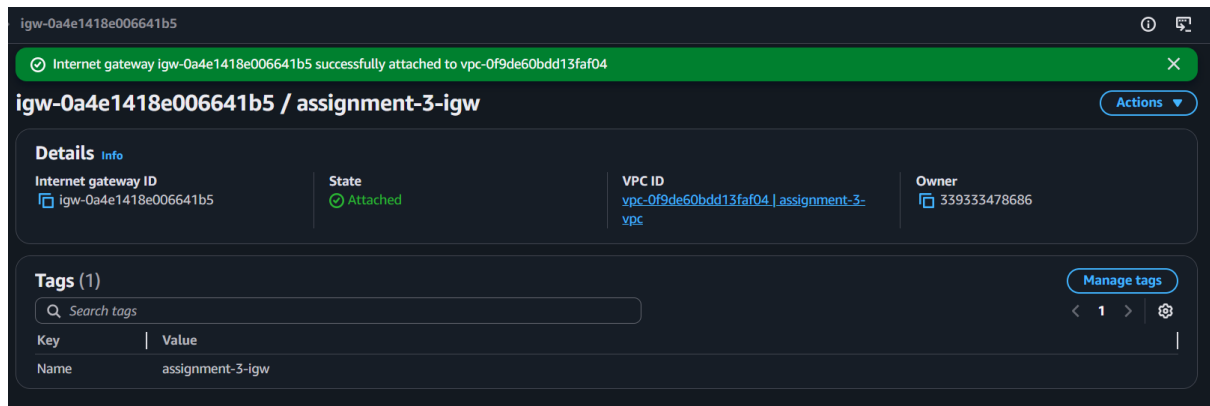
The screenshot displays the AWS Management Console interface for a subnet. At the top, a green banner indicates the successful creation of subnet-02625ba6d7756526b. Below this, the 'Subnets (1/1)' section shows a table with one entry: 'private-subnet-az2' with ID 'subnet-02625ba6d7756526b', state 'Available', VPC 'vpc-0f9de60bdd13faf04', and IPv4 CIDR '10.0.12.0/24'. The details panel for this subnet shows the following information:

- Subnet ID:** subnet-02625ba6d7756526b
- Subnet ARN:** arn:aws:ec2:ap-south-1:3393334786:86:subnet/subnet-02625ba6d7756526b
- State:** Available
- Block Public Access:** Off
- IPv4 CIDR:** 10.0.12.0/24
- Available IPv4 addresses:** 251
- IPv6 CIDR:** -
- Availability Zone:** ap-south-1 (ap-south-1b)
- Network border group:** ap-south-1
- VPC:** vpc-0f9de60bdd13faf04 | assignment-3-vpc
- Route table:** -
- Network ACL:** -
- Default subnet:** No
- Auto-assign public IPv4 address:** No
- Auto-assign customer-owned IPv4 address:** No
- Customer-owned IPv4 pool:** -
- Auto-assign IPv6 address:** No
- IPv4 CIDR reservations:** -
- Resource name DNS A record:** -

Public subnets are used for internet-accessible resources while private subnets are used for secure internal resources.

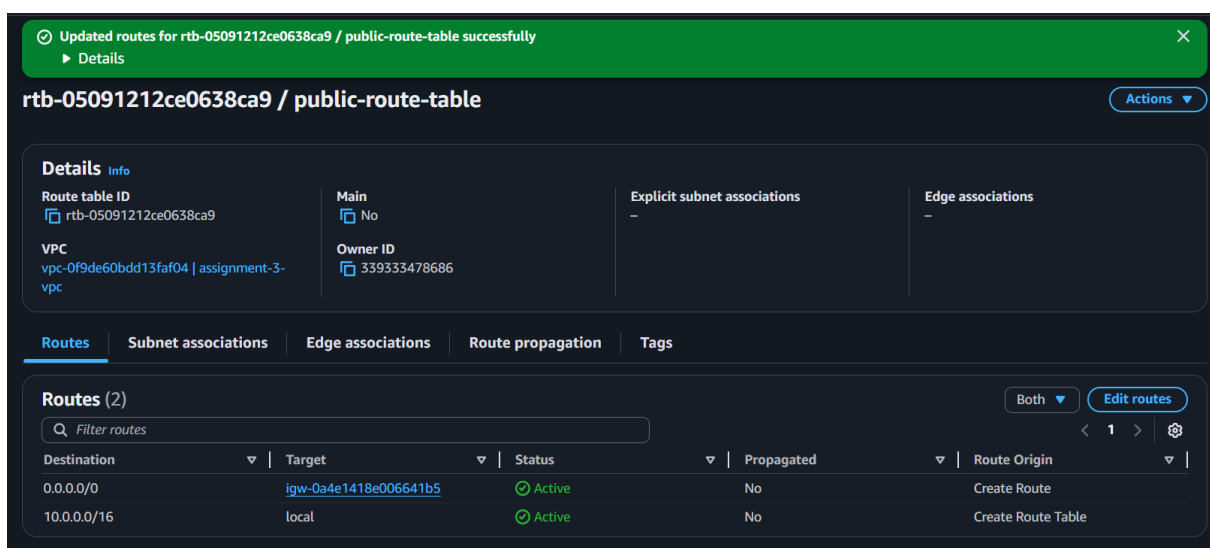
### Step 3: Create and Attach Internet Gateway

- An Internet Gateway named assignment-3-igw was created and attached to the VPC.
- The Internet Gateway allows public subnet resources to communicate with the internet.



### Step 4: Create Public Route Table

- A public route table named public-route-table was created.
- A route 0.0.0.0/0 was added and pointed to the Internet Gateway to allow internet access for public subnets.



## Step 5: Associate Public Subnets

- The public subnets were associated with the public route table.
- This association enables internet connectivity for resources inside the public subnets.

The screenshot shows the AWS Management Console interface for Route Tables. At the top, there's a header for 'Route tables (1/5)' with a search bar and a table of route tables. The table has columns: Name, Route table ID, Explicit subnet associations, Edge associations, Main, and VPC. The 'public-route-table' is selected and highlighted. Below the table, the details for 'rtb-05091212ce0638ca9 / public-route-table' are shown. The 'Details' tab is active, displaying the Route table ID, VPC, Main status (No), and Owner ID. A tooltip is visible over the 'Explicit subnet associations' field, showing two subnets: 'subnet-0e1d1ec8939974380 / public-subnet-az2' and 'subnet-0eb710d22440747a7 / public-subnet-az1'.

| Name               | Route table ID        | Explicit subnet associations | Edge associations | Main | VPC                          |
|--------------------|-----------------------|------------------------------|-------------------|------|------------------------------|
| -                  | rtb-0cc74e8a63ed00360 | -                            | -                 | Yes  | vpc-01eaa90696ea8e9a9        |
| RDS-Pvt-rt         | rtb-089f6ba0d3d63bfb8 | 3 subnets                    | -                 | No   | vpc-01eaa90696ea8e9a9        |
| -                  | rtb-0f5e20220d2c99d66 | -                            | -                 | Yes  | vpc-0f9de60bdd13faf04   assi |
| public-route-table | rtb-05091212ce0638ca9 | 2 subnets                    | -                 | No   | vpc-0f9de60bdd13faf04   assi |

**rtb-05091212ce0638ca9 / public-route-table**

**Details**

Route table ID: rtb-05091212ce0638ca9

VPC: vpc-0f9de60bdd13faf04 | assignment-3-vpc

Main: No

Owner ID: 339333478686

Explicit subnet associations: 2 subnets

- subnet-0e1d1ec8939974380 / public-subnet-az2
- subnet-0eb710d22440747a7 / public-subnet-az1

## Step 6: Create Private Route Table

- A private route table named private-route-table was created.
- Only local routes were configured and no internet gateway route was added to keep private subnets isolated from the internet.

The screenshot shows the AWS Management Console interface for a newly created private route table. A green notification banner at the top states: 'Route table rtb-0fbc7510c12c18345 | private-route-table was created successfully.' Below this, the header for 'rtb-0fbc7510c12c18345 / private-route-table' is shown. The 'Details' tab is active, displaying the Route table ID, VPC, Main status (No), and Owner ID. The 'Routes' tab is also visible, showing a single route with the destination '10.0.0.0/16', target 'local', status 'Active', and route origin 'Create Route Table'.

**Route table rtb-0fbc7510c12c18345 | private-route-table was created successfully.**

**rtb-0fbc7510c12c18345 / private-route-table**

**Details**

Route table ID: rtb-0fbc7510c12c18345

VPC: vpc-0f9de60bdd13faf04 | assignment-3-vpc

Main: No

Owner ID: 339333478686

Explicit subnet associations: -

Edge associations: -

**Routes**

Routes (1)

Filter routes

| Destination | Target | Status | Propagated | Route Origin       |
|-------------|--------|--------|------------|--------------------|
| 10.0.0.0/16 | local  | Active | No         | Create Route Table |

## Step 7: Associate Private Subnets

- The private subnets were associated with the private route table.
- These subnets can communicate only within the VPC and are not directly accessible from the internet.

**rtb-0fbc7510c12c18345 / private-route-table**

**Details** Info

Route table ID: rtb-0fbc7510c12c18345

VPC: vpc-0f9de60bdd13faf04 | assignment-3-vpc

Main: No

Owner ID: 339333478686

Explicit subnets: 2 subnets

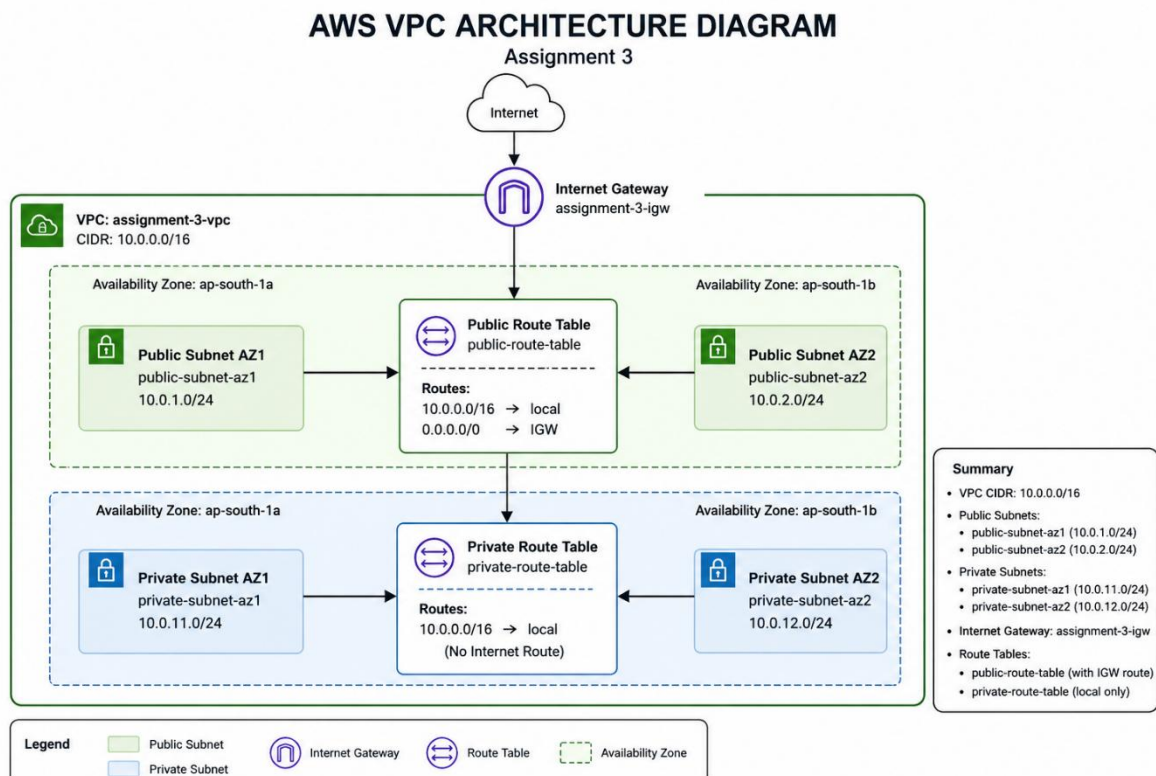
- subnet-02625ba6d7756526b / private-subnet-az2
- subnet-0ada64a7923c704de / private-subnet-az1

**Routes (1)**

| Destination | Target | Status | Propagated | Route Origin       |
|-------------|--------|--------|------------|--------------------|
| 10.0.0.0/16 | local  | Active | No         | Create Route Table |

## Step 8: Architecture Diagram

The architecture diagram below represents the complete VPC network design including public subnets, private subnets, route tables, and internet gateway configuration.



**Key Learnings:**

- Learned VPC networking concepts
- Understood public and private subnet architecture
- Configured Internet Gateway and route tables
- Learned subnet association concepts
- Improved understanding of AWS cloud networking

**Conclusion:**

Successfully designed and configured a custom AWS VPC with public and private subnet architecture. Internet Gateway, route tables, and subnet associations were configured to control internet access and internal network communication.